

Bachelor of Science in Software Engineering

[Home](#) / [Academics](#) / [Academic Programmes](#) / [Undergraduate Programs](#) / [Bachelor of Science in Software Engineering](#)

Programme Name and Duration

The programme name is Bachelor of Science in Software Engineering (BSSE) and is hosted at the Department of Networks. The programme runs for 4 academic years comprising of four (8) semesters.

Programme Description

The programme equips a Software Engineering Graduate with an adequate grasp of the principles and techniques required to produce software systems on time, within budget and with few or no defects. Our graduate is expected to apply these principles of engineering to the design, development, maintaining, testing, and evaluation of the software and systems that make computers or anything containing software work. Thus, the training given in our program ensures that our graduates have a personal, business and technical skill to advance their career wherever they want to go. In particular, we prepare them to understand that choosing an occupation, getting that first graduate job, and growing in that job will require planning, self awareness, flexibility and a forward-looking attitude to help manage the personal transitions, as well as the technological and economic changes of the future. In that regard, BSSE program offers diverse career options as including:

1. Software developer/ engineer,
2. Software architect,
3. Analyst/ programmer,
4. Games developer,
5. Project manager,
6. Mobile application developer,
7. Test analyst/engineer

Target Group

The programme targets three categories of people, namely A level certificate holders and Diploma holders and Degree holders in relevant programmes.

The programme targets to enroll 100 students on day and 100 students on evening and hence a total of 200 students.

Nature of the Programme

This programme cater is a day programme for all government sponsored studdents and caters for both day and evening private sponsored students.

Programme Duration

The duration for the BSc. in Software Engineering degree programme is four

(4) academic years comprising 8 semesters, two practical skiling recess terms in first and second year, and a field attachment session in third year.

Objectives	Programme Outcomes	Career Options	Entry Requirements
The general objective of the programme is to train/skill student the theory and practices of; design, development, securing and maintenance of software. Other objectives of the BSSE programme are: • To build human resource capacity that is proficient in software design and development in a variety of programming languages and platforms using a methodical approach for both the public and private sectors. Such human resource that is well grounded in the fundamental concepts of software engineering and able to continue their professional development throughout their • To equip graduates with lifelong learning skills capable of functioning responsibly in diverse environments and able to work in • To empower graduates with ability of creating their own jobs in software engineering • To foster interdisciplinary studies which enable the application of software knowledge in non-computating disciplines.			

Programme Structure

Programme Structure										
CODE	COURSE NAME	CU	LH	PH	TH	CH	Type	Remark	Origin	
Year 1 – Semester I (5 Core Courses, No Electives) – Total Credit Units=19										
BSE1106	Problem Solving and Programming Concepts	4	30	30	30	60	Core	Modifie d	NW	
CSK1101	Communication Skills	4	30	60		60	Core	Old	Langu	
BSE1107	Mathematics for Software Engineers	3	30	–	30	45	Core	New	NW	
BSE1108	Technical Analysis and Design	4	30	30	30	60	Core	New	NW	
IST1101	Foundations of Information Systems & Technology	4	45	30	–	60	Core	Old	IS	
	Audit Course									
CSC	Computer Literacy	4	30	30	30	60	AU	Old	CS	
Year 1 – Semester II (5 Core Courses, No Electives) – Total Credit Units=18										
BSE1206	Software Development Principles	3	45	–	–	45	Core	Old	NW	
MTH2203	Numerical Analysis I	3	30	–	30	45	Core	Old	Math	
BSE1209	Object Oriented Programming I	4	30	30	30	60	Core	New	NW	
IST1203	Data and Information Management I	4	30	60	–	60	Core	Old	IS	
BSE1208	Introduction to Web Development	4	30	30	30	60	Core	New	NW	
Year I Recess (1 Core Course, No Electives) – Total Credit Units = 5										
BSE1302	Software Engineering Practical Skills Project I	5	–	150	–	75	Core	Old	NW	
Year 2 – Semester I (5 Core Courses, No Electives) – Total Credit Units=19										
CSC2114	Artificial Intelligence	4	30	30	30	60	Core	Modifie	CS	
CSC2100	Data Structures and Algorithms	4	30	30	30	60	Core	Old	CS	
BSE2106	Computer Networks	4	30	30	30	60	Core	Old	NW	
BSE2105	Formal Methods	3	45	–	–	45	Core	Old	NW	
BSE2107	Object Oriented Programming II	4	30	30	30	60	Core	New	NW	
Year 2 – Semester II (5 Core Courses, No Elective) – Total Credit Units=19										
CSC2200	Operating Systems	4	30	30	30	60	Core	Modified	CS	
BSE2206	Data Communication	4	45	30	30	60	Core	Old	NW	
BSE2207	Emerging Web Development Technologies	4	30	30	30	60	Core	New	NW	
BSE2208	Requirements Engineering	3	45	–	–	45	Core	Modified	NW	
BSE2209	Mobile Programming Project	4	15	60	30	60	Core	New	NW	
Year II Recess(1 Core Course, No Electives) – Total Credit Units = 5										
BSE2302	Software Engineering Practical Skills Project II	5	–	150	–	75	Core	Old	NW	
Year 3 – Semester I (5 Core Courses) – Total Credit Units=19										
BSE3114	Internet of Things: Technologies & Protocols	4	30	60	–	60	Core	Old	NW	
BSE3113	Embedded Systems I	4	30	30	30	60	Core	New	NW	
BSE3104	Software Metrics	3	45	–	–	45	Core	Old	NW	
CSC3119	User Interface Design	4	30	30	30	60	Core	Modified	CS	
	Electives									
BSE3106	Mobile Networks and Computing	4	45	30	–	60	Elective	old	NW	
BSE3105	Software Evolution	4	45	–	30	60	Elective	Old	NW	
Year 3 – Semester II (5 Core Courses, Choose ONLY 1 Elective) – Total Credit Units=18										
BSE3210	Software Architecture and Patterns	3	45	–	–	45	Core	New	NW	
BSE3211	Software Testing and Verification	3	45	–	–	45	Core	New	NW	
IST2203	Research Methodology	4	45	–	30	60	Core	Old	IT	
CSC2206	Machine Learning	4	30	30	30	60	Core	Old	CS	
	Electives									
BSE3214	Cloud Computing and Big Data	4	30	30	30	60	Elective	New	NW	
BSE3213	Embedded Systems II	4	30	30	30	60	Elective	New	NW	
Year III Recess(1 Core Course, No Electives) – Total Credit Units = 5										
BSE3302	Field Attachment	5	–	150	–	75	Core	Old	NW	
Year 4 – Semester I (4 Core Courses, No Electives)- Total Credit Units=15										
BSE4100	Software Engineering Project I	5	–	150	–	75	Core	Old	NW	
BSE4106	ICT Innovation and Entrepreneurship	3	45	–	–	45	Core	New	NW	
BSE4104	Emerging Trends in Software Engineering	3	45	–	–	45	Core	Modified	NW	
BSE4105	Software Integration and Deployment	4	30	60	–	60	Core	New	NW	
Year 4 – Semester II (4 Core Courses, No Electives)- Total Credit Units=15										
BSE4200	Software Engineering Project II	5	–	150	–	75	Core	Old	NW	
BSE4202	Software Security	4	30	60	–	60	Core	Modified	NW	
BSE4203	Software Engineering Standards & Ethics	3	45	–	–	45	Core	New	NW	
BSE4205	Software Quality Management	3	45	–	–	45	Core	New	NW	

HOW TO APPLY

FEES STRUCTURE

APPLY ONLINE



Makerere University College of Computing and Information Science (Mak -CoCIS) is one of the nine (9) colleges of Makerere University.

Feature Links

- › [Makerere University](#)
- › [MUELE \(E-Learning\)](#)
- › [Academic Registrar](#)
- › [Makerere University News](#)
- › [Makerere University@100](#)

Support

- › [Academic Calendar](#)
- › [Fees Structure](#)
- › [Timetables](#)
- › [Admissions](#)
- › [FAQs](#)

Contact Us

P.O.Box 7062, Kampala, Uganda.
 principal.cis@mak.ac.ug / pr.cocis@mak.ac.ug
 +256 414 540628/534560/1-97
 +256 414 540620

